

Billion-Scale Vector Search

IVF-PQ ANN engine from scratch (NumPy)

November 7, 2023

Headline Result

recall@10 0.75-0.92 vs exact; ~7x compression; 1B
= ~67 GiB

Prepared for: Sai Veda

Publishing account: Nikeshk834

Project Context

Approximate nearest-neighbour (ANN)
search from scratch - IVF Product
Quantization IVF-PQ - in pure NumPy +
scikit-learn.

This PDF dossier is intentionally visual and interview-ready: it captures the business problem, the technical architecture, the validation evidence, and the committed repository artifacts in a format that reads like a delivered project brief.

Repository slug

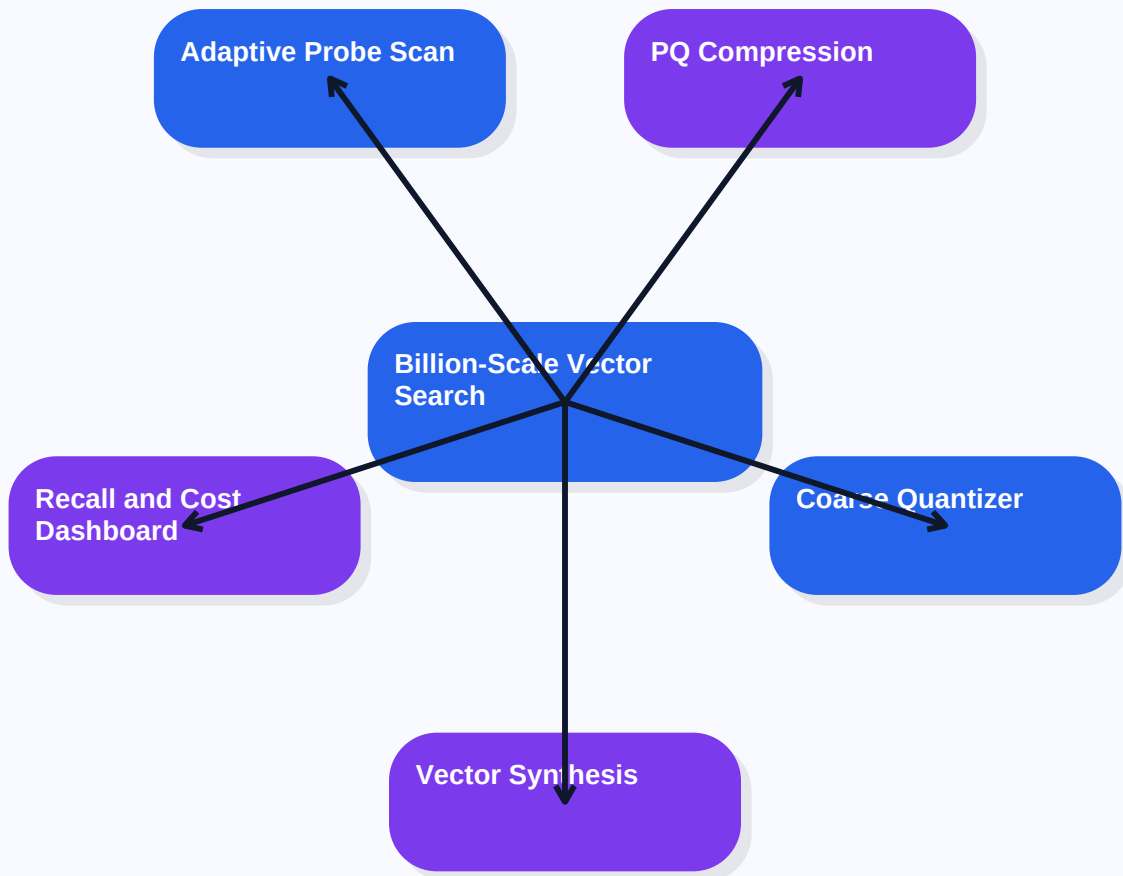
02-billion-scale-vector-search

Validation index

22 tests

Architecture Storyboard

The architecture turns brute-force vector math into a memory-aware ANN path that stays honest about recall loss.



High-level focus area: The architecture turns brute-force vector math into a memory-aware ANN path that stays honest about recall loss.

Validation And Visual Evidence

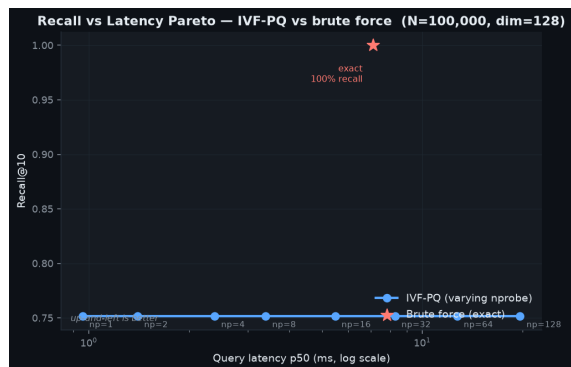
recall@10 0.75-0.92 vs exact; ~7x compression; 1B = ~67 GiB

22 passing tests

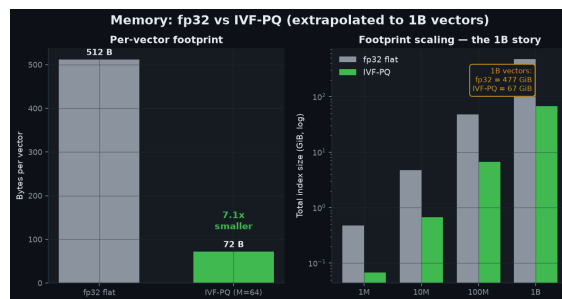
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Selected benchmark table

N	nlist	build (s)	recall@10	p50 (ms)
10,000	200	35.233	0.9247	3.3271
100,000	632	70.128	0.7514	10.8552
1,000,000	2000	341.957	0.8744	10.4025



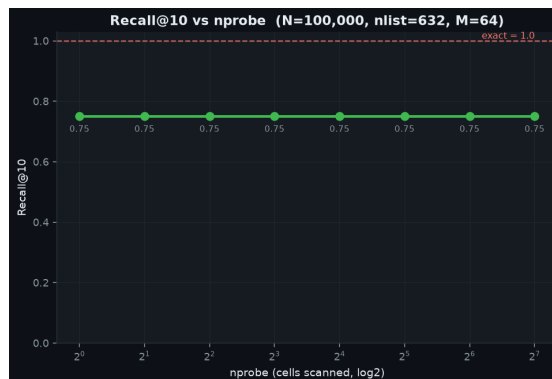
Recall vs Latency Pareto - IVF-PQ vs brute force



Memory footprint: fp32 vs IVF-PQ, extrapolated to 1B



Index build-time scaling (log-log)



Recall@10 vs nprobe (the accuracy dial)

Repository Handoff

Prepared for: Sal Veda

Publishing account: Nikeshk834

Project date: November 7, 2023

Repository: 02-billion-scale-vector-search

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from scratch - IVF Product Quantization IVF-PQ -
in pure NumPy + scikit-learn.

Committed deliverables

- README.md - product overview, quickstart, and measured headline numbers
- ARCHITECTURE.md - detailed design and scale trade-offs
- tests/ - automated validation suite
- benchmarks/ - measured run artifacts and result tables
- assets/ - generated dashboards, charts, and screenshots
- PROJECT_DOCUMENT.pdf - styled project brief for sharing and review

This document is intentionally more visual than the repository Markdown. It is meant to read like a polished project brief: business framing first, architecture second, proof third, and delivery scope last.